

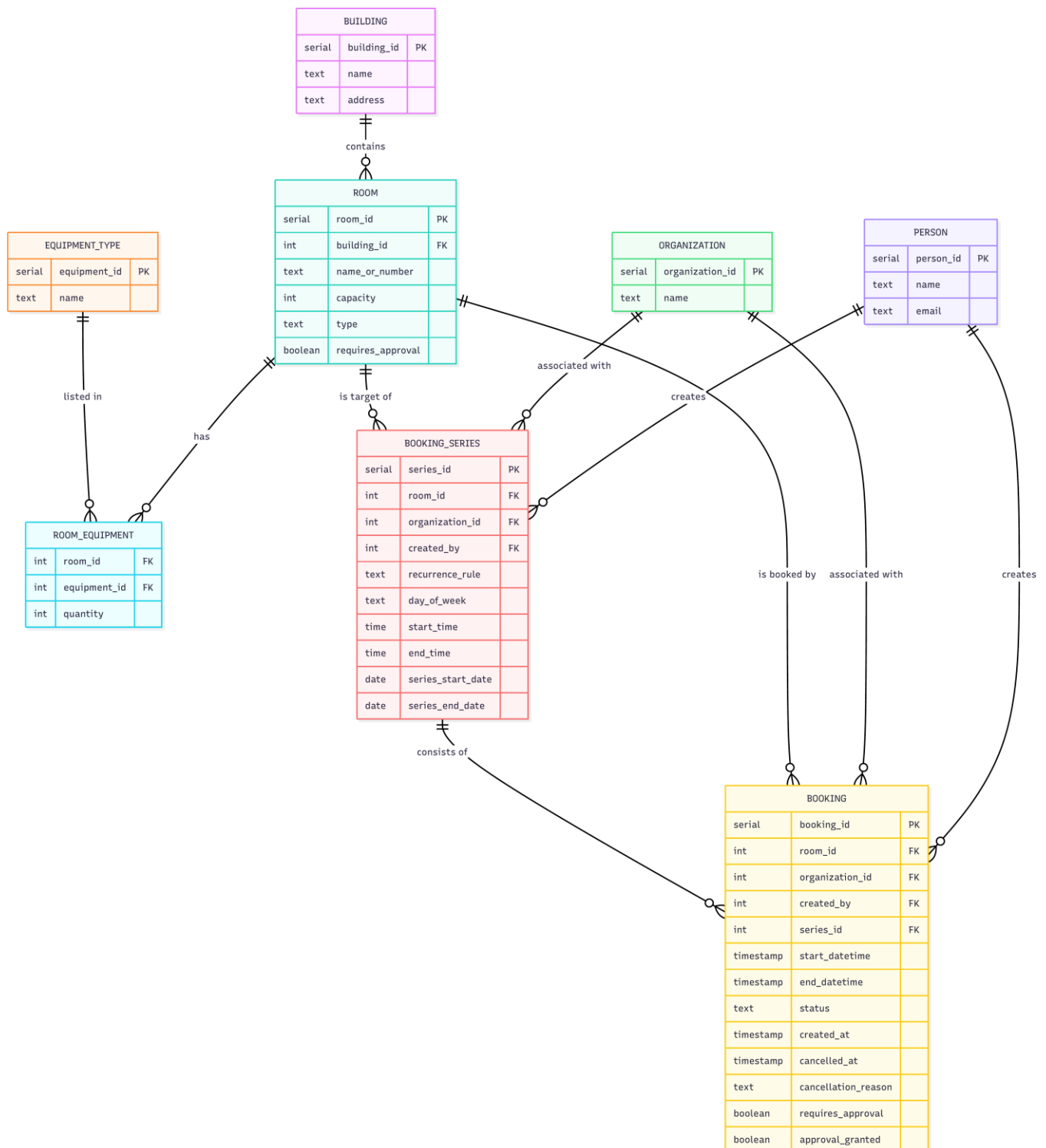
Conceptual Model & Relational Schema

Conceptual Model

ER Diagram

The diagram below is drawn in crow's foot notation. Each relationship line is read as:

- || : exactly one
- o{ : zero or many
- || - o{ : one to many (mandatory on the left, optional-many on the right)



Entities

Entity	Description
BUILDING	A university building that contains rooms
ROOM	A bookable room within a building
EQUIPMENT_TYPE	A type of equipment that may be present in rooms (e.g. projector)
ROOM_EQUIPMENT	Linking table, records which equipment types a room has, and quantity
ORGANIZATION	A student organization that makes bookings
PERSON	The individual who creates a booking or series
BOOKING_SERIES	The planning intent for a recurring booking (e.g. weekly Wednesday evenings)
BOOKING	One time slot, either a standalone booking or one occurrence of a series

Relationships and Cardinalities

Relationship	Cardinality	Description
BUILDING -> ROOM	1 to many	A building contains zero or more rooms; a room belongs to exactly one building
ROOM -> ROOM_EQUIPMENT	1 to many	A room has zero or more equipment entries
EQUIPMENT_TYPE -> ROOM_EQUIPMENT	1 to many	An equipment type appears in zero or more rooms
PERSON -> BOOKING	1 to many	A person creates zero or more bookings
PERSON -> BOOKING_SERIES	1 to many	A person creates zero or more booking series
ORGANIZATION -> BOOKING	0/1 to many	A booking may be associated with zero or one organization (nullable); an organization has zero or more bookings
ORGANIZATION -> BOOKING_SERIES	0/1 to many	A series may be associated with zero or one organization (nullable); an organization has zero or more series
ROOM -> BOOKING	1 to many	A room is the target of zero or more bookings
ROOM -> BOOKING_SERIES	1 to many	A room is the target of zero or more series
BOOKING_SERIES -> BOOKING	1 to many	A series consists of zero or more individual booking occurrences; a standalone booking has no series

Main Attributes per Entity

BUILDING

- building_id - surrogate primary key

- name - building name
- address - physical address

ROOM

- room_id - surrogate primary key
- building_id - FK to BUILDING
- name_or_number - room identifier within the building
- capacity - maximum number of people (must be > 0)
- type - one of: meeting room, lecture room, studio, workshop room
- requires_approval - whether the room requires special access approval (current policy)

EQUIPMENT_TYPE

- equipment_id - surrogate primary key
- name - equipment name (unique); e.g. projector, whiteboard, microphone, video camera

ROOM_EQUIPMENT (linking table)

- room_id - FK to ROOM (part of composite PK)
- equipment_id - FK to EQUIPMENT_TYPE (part of composite PK)
- quantity - how many units of this equipment type the room has (must be > 0)

ORGANIZATION

- organization_id - surrogate primary key
- name - organization name (unique)

PERSON

- person_id - surrogate primary key
- name - full name
- email - contact email (unique); only personal data stored

BOOKING_SERIES

- series_id - surrogate primary key
- room_id - FK to ROOM
- organization_id - FK to ORGANIZATION; nullable (NULL if created by an individual)
- created_by - FK to PERSON
- recurrence_rule - descriptive rule string (e.g. "weekly", "biweekly")
- day_of_week - one of: Monday through Sunday
- start_time - time of day the slot starts
- end_time - time of day the slot ends (must be after start_time)
- series_start_date - date of first occurrence
- series_end_date - date of last occurrence (must be $\geq$ series_start_date)

BOOKING

- booking_id - surrogate primary key
- room_id - FK to ROOM
- organization_id - FK to ORGANIZATION; nullable (NULL if booking made by an individual)
- created_by - FK to PERSON
- series_id - FK to BOOKING_SERIES; nullable - NULL for standalone bookings
- start_datetime - start of the booking slot
- end_datetime - end of the booking slot (must be after start_datetime; same calendar day)
- status - one of: pending, confirmed, cancelled, rejected
- created_at - timestamp when the booking was created
- cancelled_at - nullable; timestamp of cancellation
- cancellation_reason - nullable; short free text
- requires_approval - snapshot of the room's approval policy at booking creation time

- approval_granted - nullable Boolean: NULL if not required or pending; TRUE if approved; FALSE if rejected

Section 3 - Relational Schema

Tables Overview

Table	Primary Key	Foreign Keys
building	building_id	-
room	room_id	building_id -> building
equipment_type	equipment_id	-
room_equipment	(room_id, equipment_id)	room_id -> room, equipment_id -> equipment_type
organization	organization_id	-
person	person_id	-
booking_series	series_id	room_id -> room, organization_id -> organization, created_by -> person
booking	booking_id	room_id -> room, organization_id -> organization, created_by -> person, series_id -> booking_series

Table Definitions

building

Column	Type	Nullable	Notes
building_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Building name
address	TEXT	NOT NULL	Physical address

room

Column	Type	Nullable	Notes
room_id	SERIAL	NOT NULL	Primary key
building_id	INT	NOT NULL	FK -> building
name_or_number	TEXT	NOT NULL	Room identifier
capacity	INT	NOT NULL	Must be > 0
type	TEXT	NOT NULL	meeting room / lecture room / studio / workshop room
requires_approval	BOOLEAN	NOT NULL	Default FALSE; current room policy

Unique constraint: (building_id, name_or_number) - no two rooms in the same building share an identifier.

equipment_type

Column	Type	Nullable	Notes
equipment_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Unique equipment type name

room_equipment

Column	Type	Nullable	Notes
room_id	INT	NOT NULL	FK -> room; part of composite PK
equipment_id	INT	NOT NULL	FK -> equipment_type; part of composite PK
quantity	INT	NOT NULL	Default 1; must be > 0

organization

Column	Type	Nullable	Notes
organization_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Unique organization name

person

Column	Type	Nullable	Notes
person_id	SERIAL	NOT NULL	Primary key
name	TEXT	NOT NULL	Full name
email	TEXT	NOT NULL	Unique; only personal field stored

booking_series

Column	Type	Nullable	Notes
series_id	SERIAL	NOT NULL	Primary key
room_id	INT	NOT NULL	FK -> room
organization_id	INT	nullable	FK -> organization; NULL if series created by an individual
created_by	INT	NOT NULL	FK -> person
recurrence_rule	TEXT	NOT NULL	e.g. "weekly", "biweekly"
day_of_week	TEXT	NOT NULL	Monday–Sunday
start_time	TIME	NOT NULL	Slot start time
end_time	TIME	NOT NULL	Slot end time; must be after start_time
series_start_date	DATE	NOT NULL	First occurrence date
series_end_date	DATE	NOT NULL	Last occurrence date; must be ≥ series_start_date

booking

Column	Type	Nullable	Notes
booking_id	SERIAL	NOT NULL	Primary key
room_id	INT	NOT NULL	FK -> room
organization_id	INT	nullable	FK -> organization; NULL if booking made by an individual
created_by	INT	NOT NULL	FK -> person
series_id	INT	nullable	FK -> booking_series; NULL if standalone
start_datetime	TIMESTAMP	NOT NULL	Slot start
end_datetime	TIMESTAMP	NOT NULL	Slot end; must be after start; same calendar day
status	TEXT	NOT NULL	pending / confirmed / cancelled / rejected; default pending
created_at	TIMESTAMP	NOT NULL	Default NOW()
cancelled_at	TIMESTAMP	nullable	NULL if not cancelled
cancellation_reason	TEXT	nullable	NULL if not cancelled
requires_approval	BOOLEAN	NOT NULL	Snapshot from room at creation time; default FALSE
approval_granted	BOOLEAN	nullable	NULL = not required or pending; TRUE = approved; FALSE = rejected

Check Constraint on Approval State

requires_approval	approval_granted	Valid statuses
FALSE	NULL	confirmed, cancelled
TRUE	NULL	pending, cancelled
TRUE	TRUE	confirmed, cancelled
TRUE	FALSE	rejected, cancelled

Note: A booking with requires_approval = FALSE cannot sit as pending: if no approval is needed, the booking is confirmed or cancelled directly. This is enforced by the CHECK constraint in schema.sql.

Additional Check Constraints

- end_datetime > start_datetime
- DATE(start_datetime) = DATE(end_datetime): a booking cannot span midnight

Linking Tables

room_equipment

Resolves the many-to-many relationship between room and equipment_type. A room can have multiple equipment types; the same equipment type can appear in multiple rooms. The composite primary key (room_id, equipment_id) ensures no duplicate entries. The quantity column gives this linking table its own structure beyond the relationship itself.

Nullable Columns Summary

Table	Nullable Columns	Reason
booking	series_id	NULL for standalone bookings
booking	organization_id	NULL if booking made by an individual with no organization
booking	cancelled_at	NULL if not cancelled
booking	cancellation_reason	NULL if not cancelled; also optional even when cancelled
booking	approval_granted	NULL if approval not required or decision not yet made
booking_series	organization_id	NULL if series created by an individual with no organization